

Carnegie Mellon University Africa

Security and New Technologies

Machine Learning for Computer Networks and Security

Assane Gueye

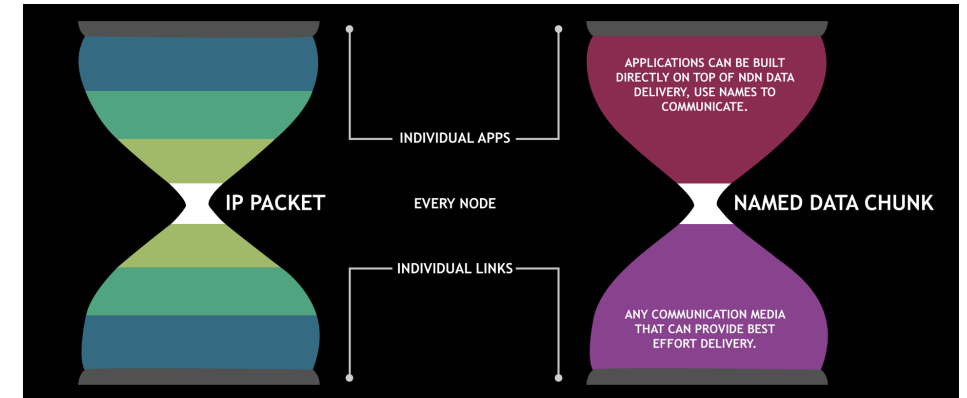
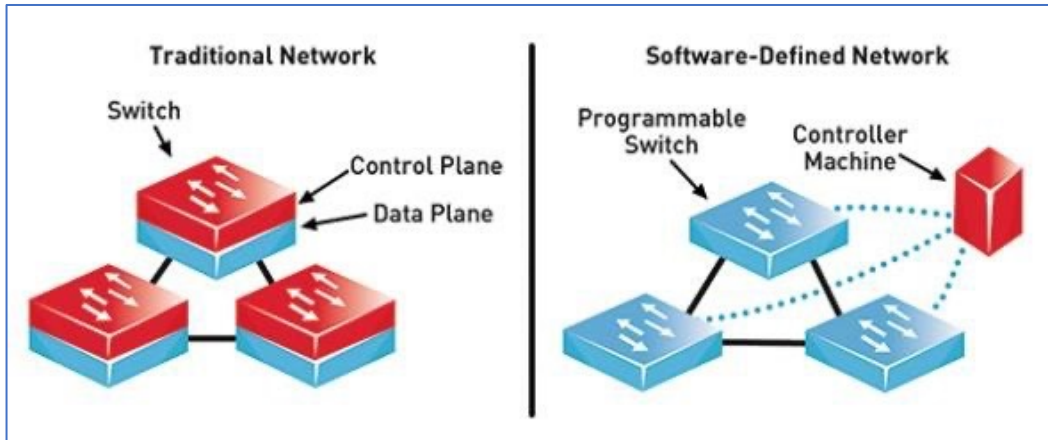
July, 2024

Courtesy to: Sylvia Ratnasamy, UC Berkeley
Junchen Jiang, U Chicago
The Internet!

Motivational Examples

Motivational Examples

NGN



Next Generation Networks are Intelligent (**NGN**):

- Self-Organization
- Self-Configuration
- Self-Optimization
- Self-Healing
- ..
- **Self-X**

Motivational Examples: ML is Networks

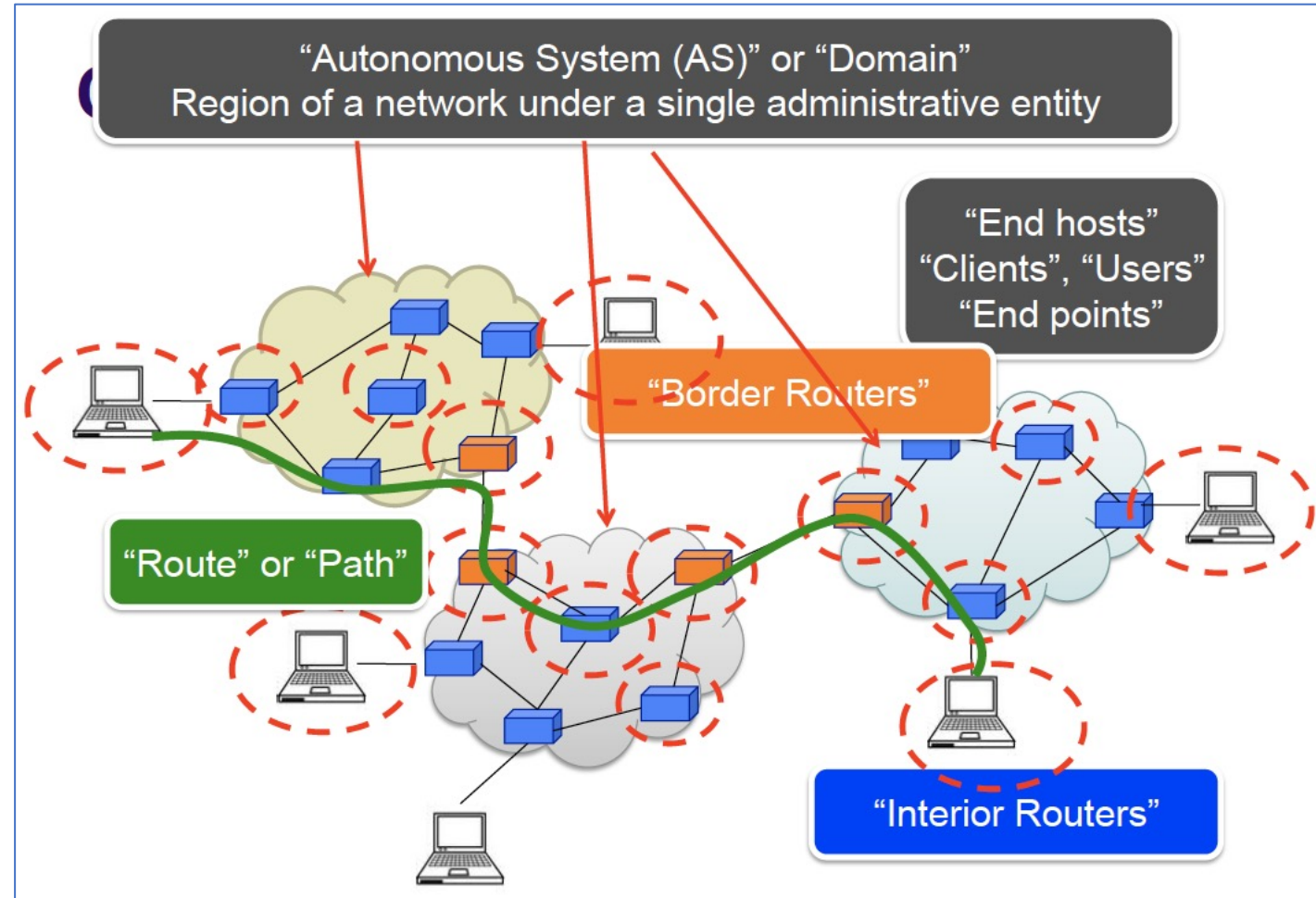
Routing

Objectives

- Correctness
- Simplicity
- Efficiency
- Robustness
- Stability
- Fairness
- Optimality

IP implementations

- Link State
- Distance Vector

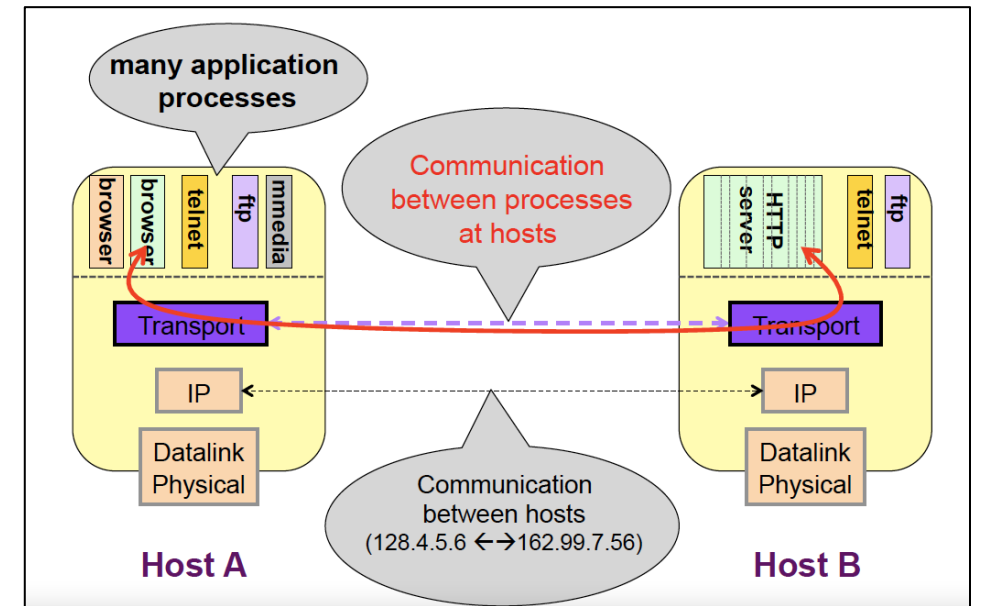


Motivational Examples

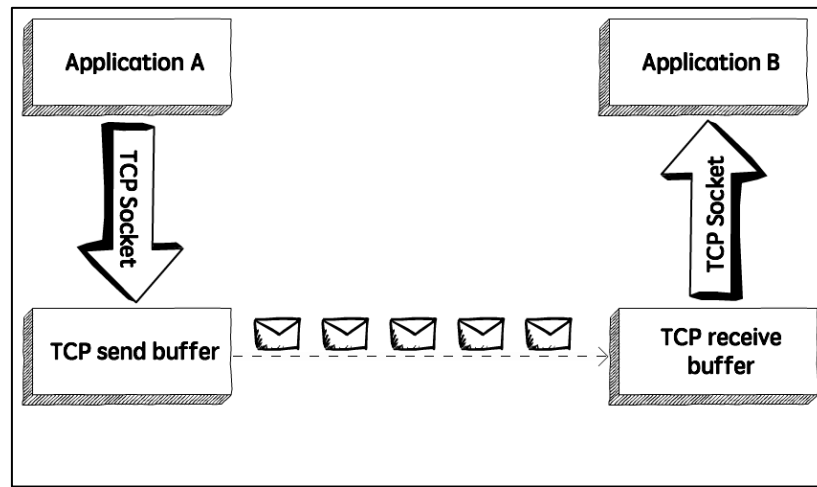
Transport and TCP

TCP: reliable communication over unreliable (IP) protocol

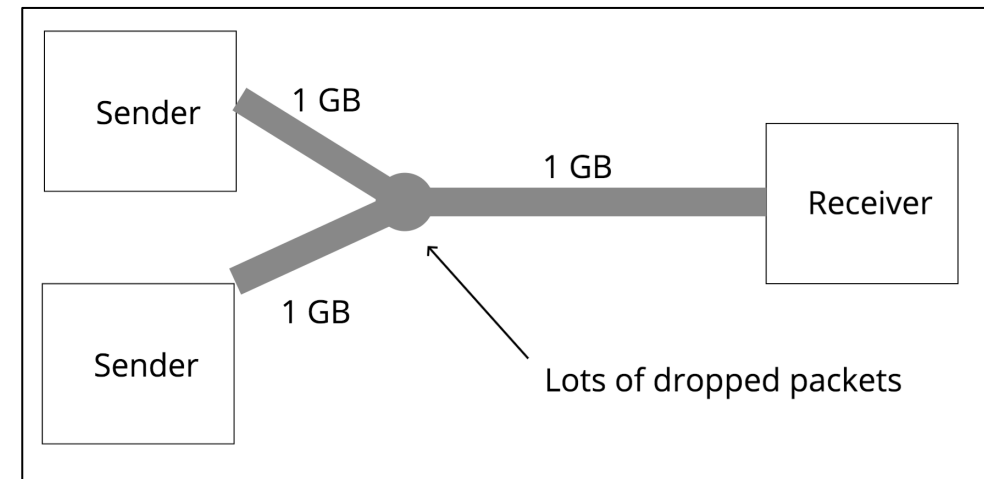
- Circuit switching
- Reliability
- Congestion control
- Flow control



Flow control



Congestion control



Motivational Examples

Traffic Classification

What and Why?

Traffic classification aims at categorization network traffic into a number of traffic classes.

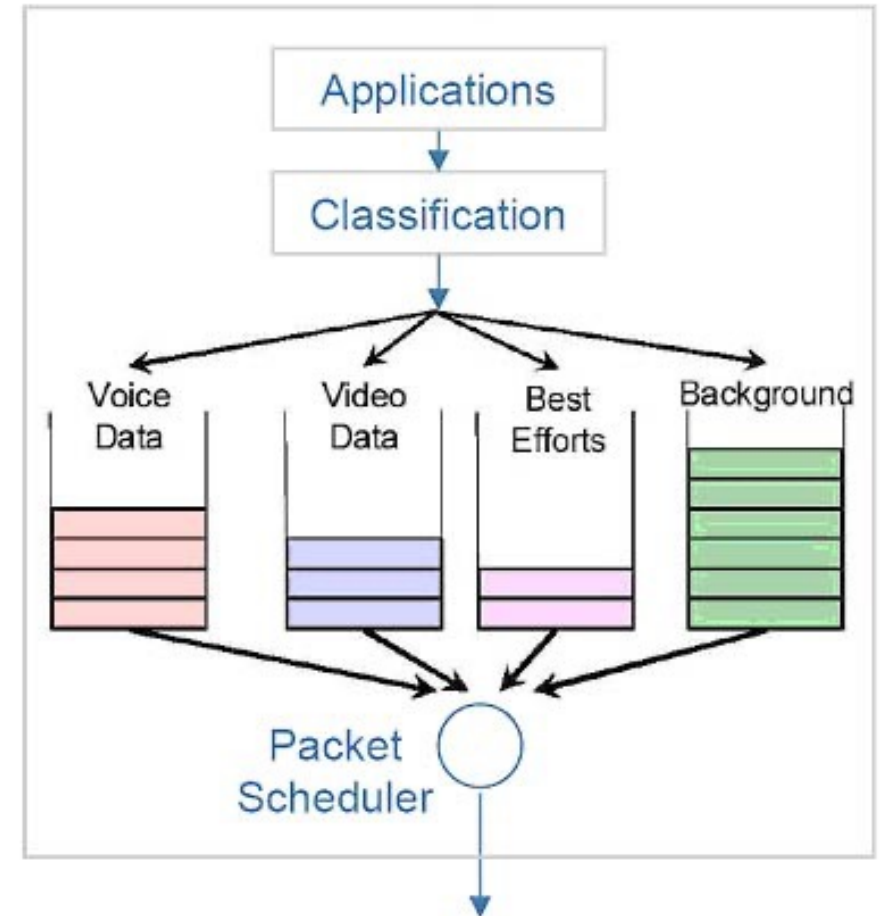
Each class/application can then be treated differently by Internet Service Providers.

Important Issues:

Battle for rights: real-time vs sharing community based on peer-to-peer traffic

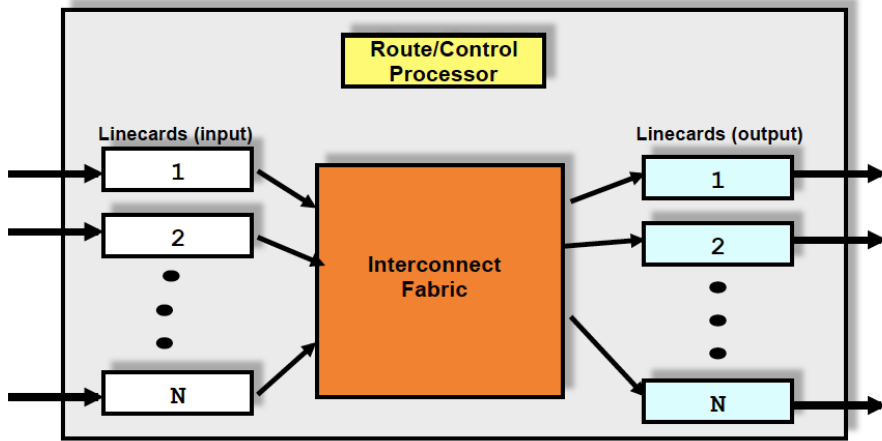
Battle for security: malicious hackers (worms, botnets) vs security management companies

Battle for **network neutrality**: ISP (e.g. AT&T, Verizon) vs content/service providers (Netflix, Youtube, P2P)



Motivational Examples

Scheduling

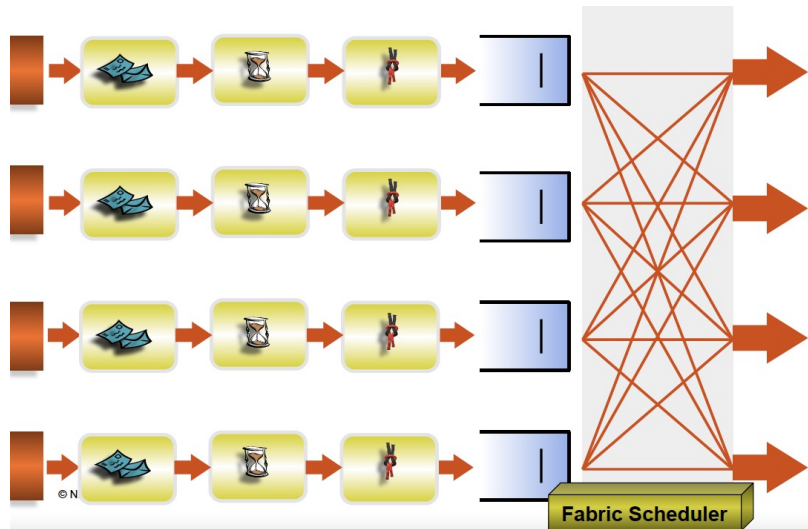


Goals:

- run links at full capacity,
- fairness across inputs

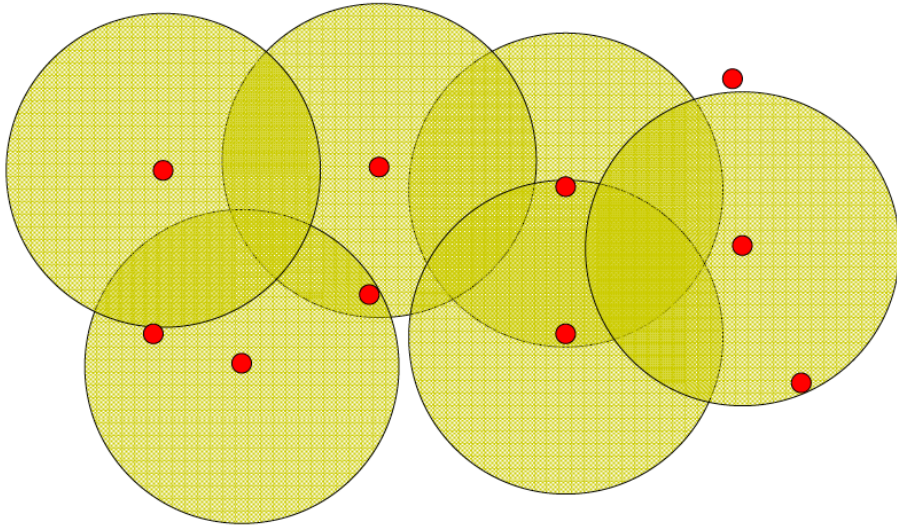
Scheduling formulated as finding a matching

Solutions look for a good maximal matching (fast)



Motivational Examples

Wireless Medium Access and Ad Hoc Networks

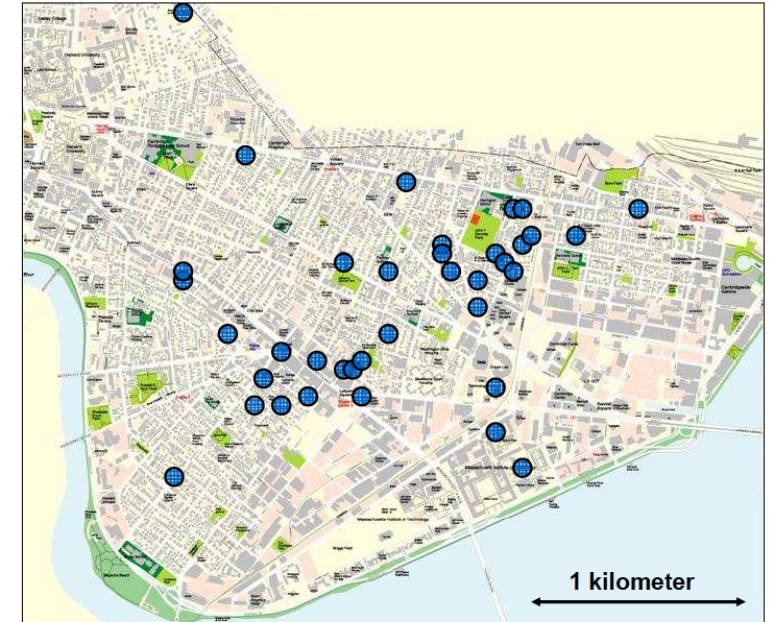


Medium Access:

- Time division
- Frequency division
- OFDM
- CSMA

Large Multihop Network

(courtesy of Sanjit Biswas, MIT)



Routing

- Mobility
- Storage

Motivational Examples

Network Management and Performance

NETWORK MANAGEMENT



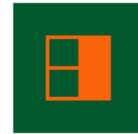
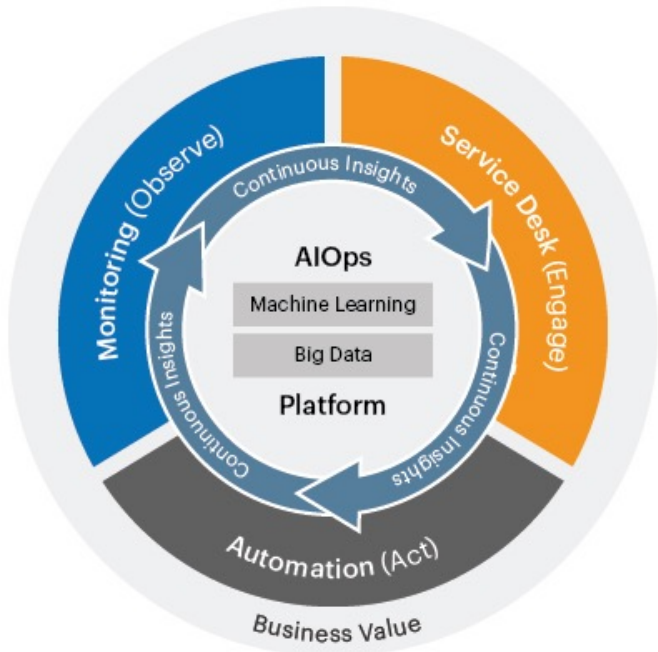
Network Performance



Bandwidth
Throughput
Delay, Delay Jitter
Route availability, Path diversity
...

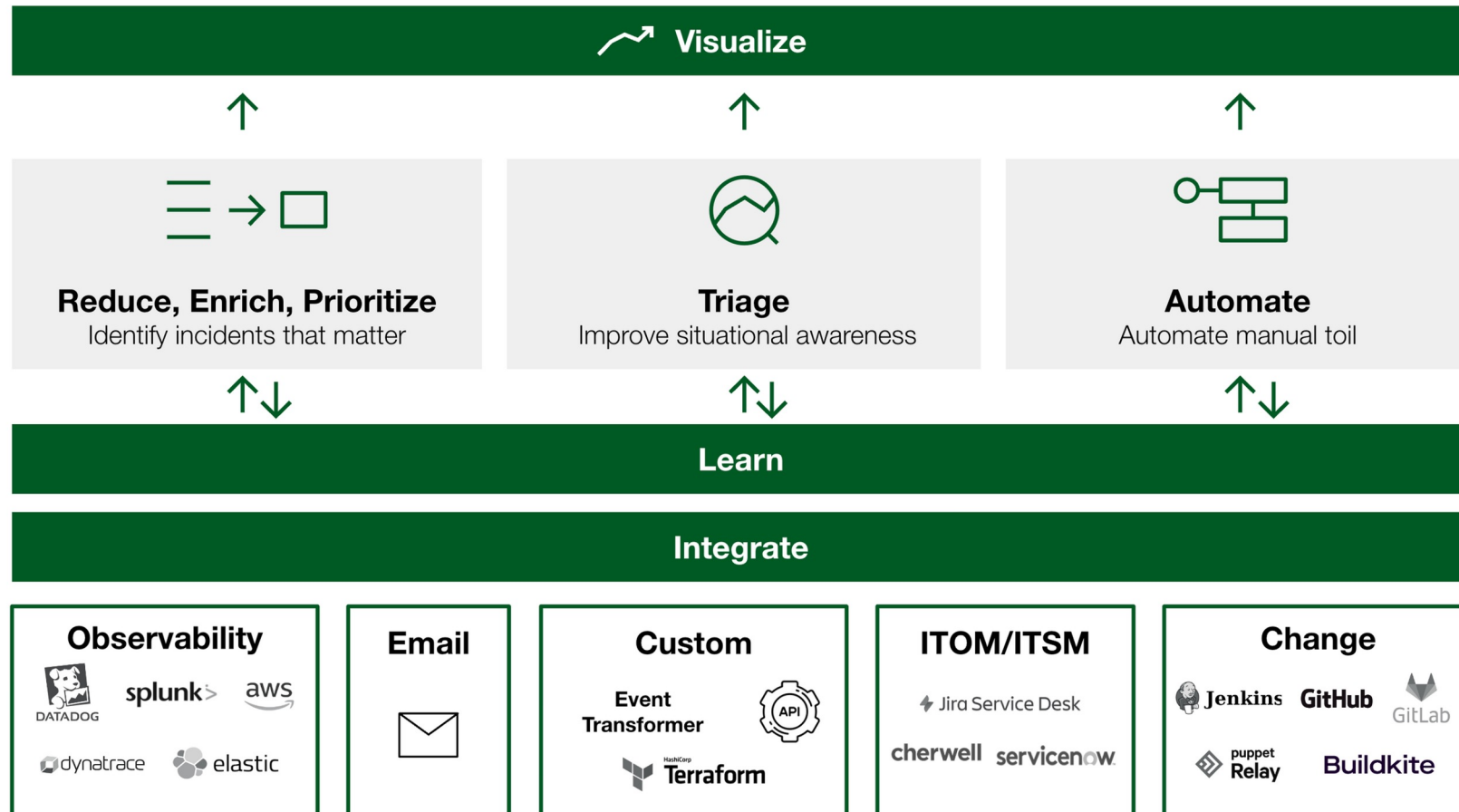
Motivational Examples

AIOps



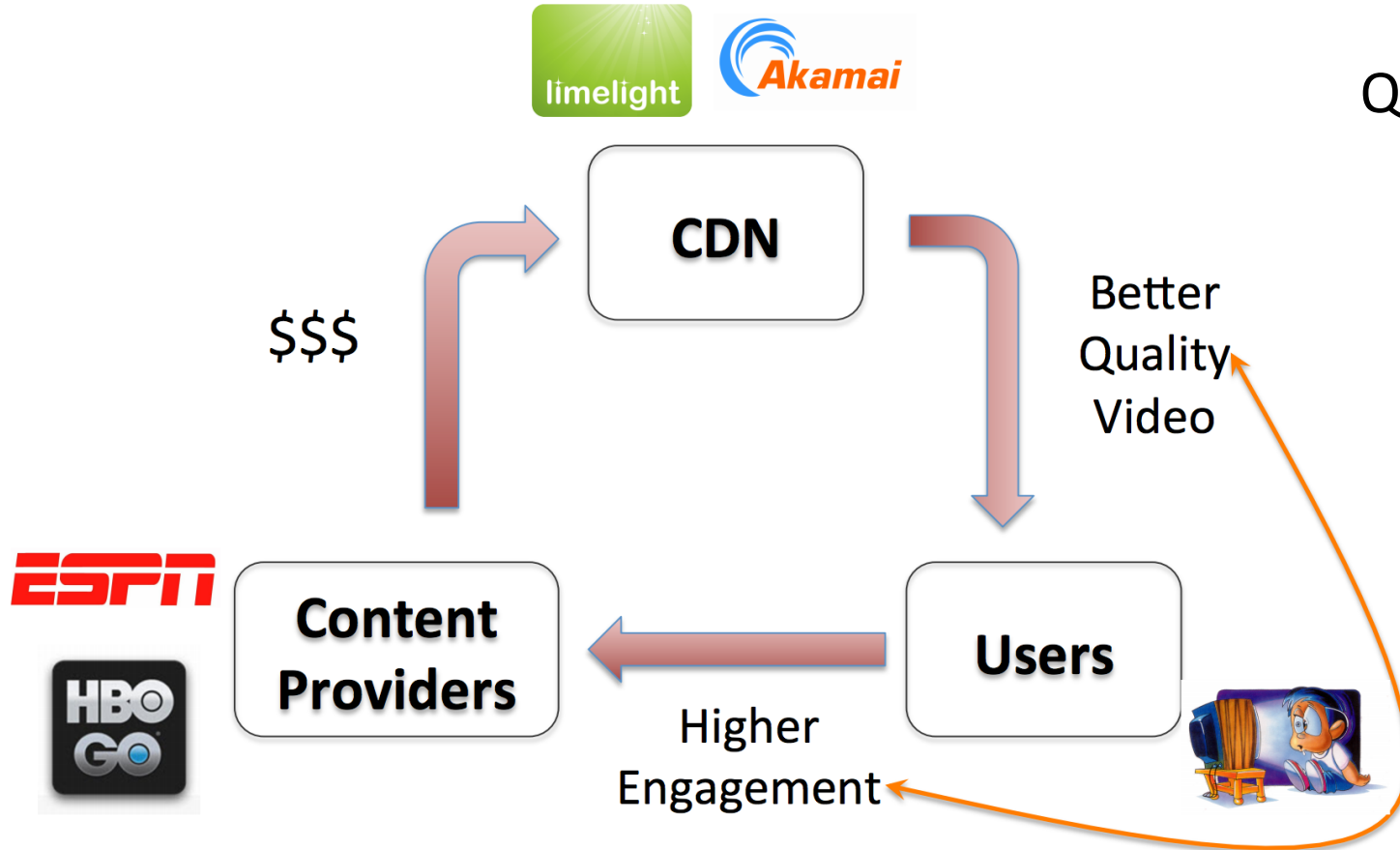
PagerDuty
AIOps

How we do it



Motivational Examples

Quality of Experience



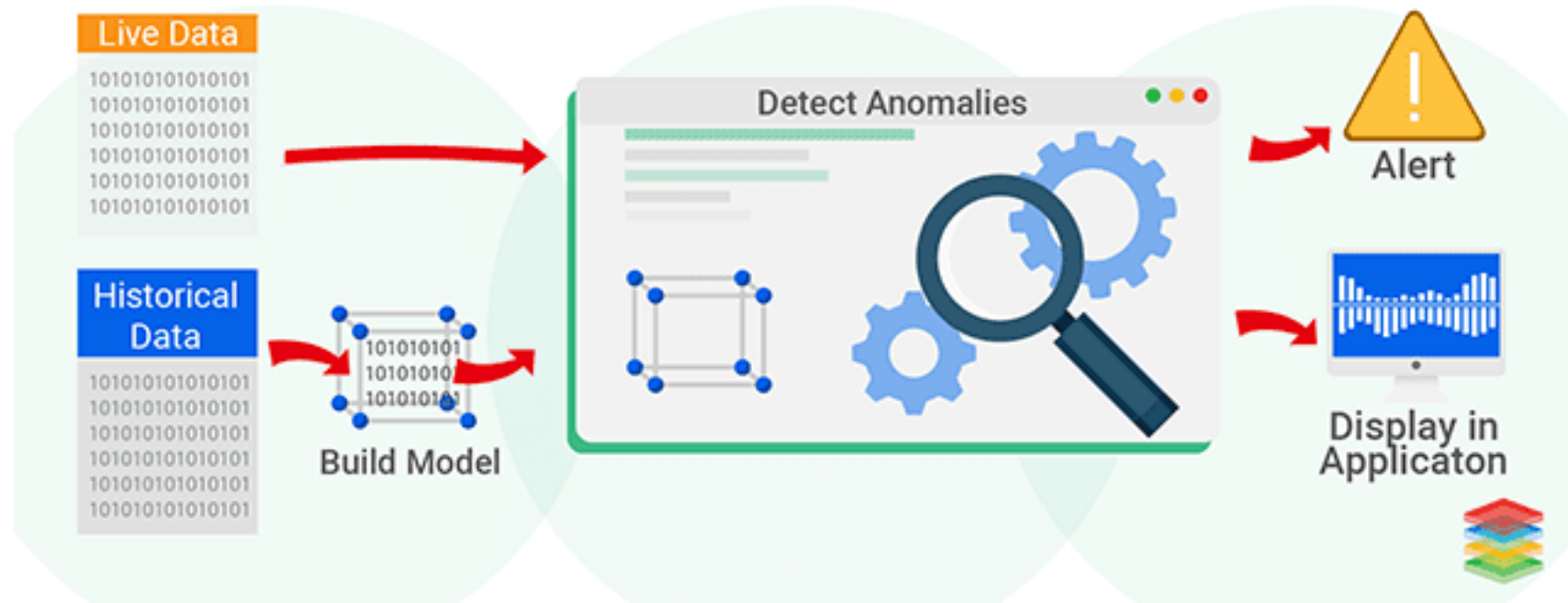
Quality of Experience (QoE) → \$\$\$

The QoE model

Motivational Examples

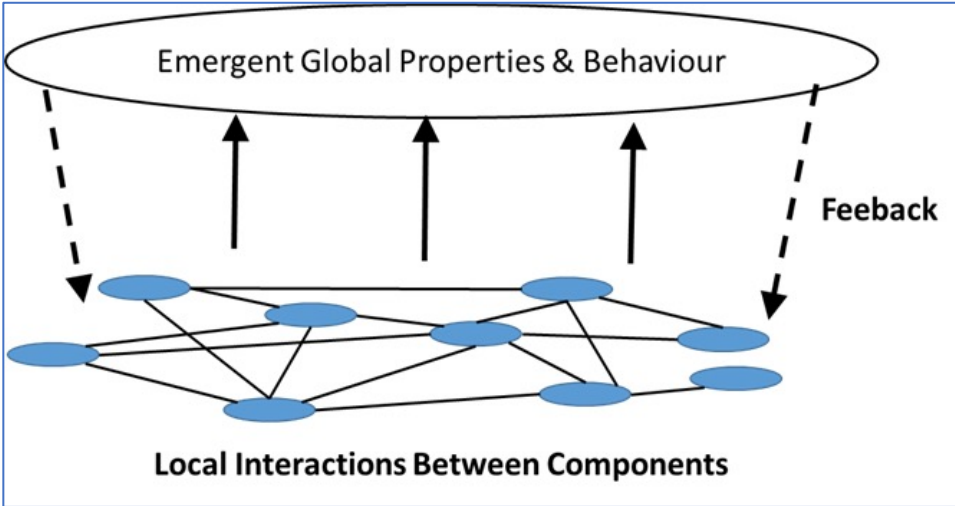
Anomaly Detection

Real Time Anomaly Detection



Motivational Examples

Emerging Behavior



Summary

- There are *various types of network data* needed for various network operations tasks.
 - E.g., telemetry, flow, packet, API, application visibility and control, application recognition data, etc.
- *Specifying the ML problem* is the most significant task in applying ML to networking and cybersecurity tasks.
 - Determines the *type of network data* that should be *collected*.
- *Data preparation* is the most critical activity in applying ML – *data collection, data exploration, feature extraction*, (and *feature engineering* where needed).
- ML can be applied in various areas of networks operations e.g., *traffic classification/identification, traffic prediction, anomaly detection, fault management, network security*, etc.
- *Datasets* for ML: Examples of *historical data sets*. But you can also collect *real-time data sets*.

Security of ML Systems

Motivational Examples

ML security

Machine learning security risks



Data confidentiality



Transfer learning attack



System manipulation



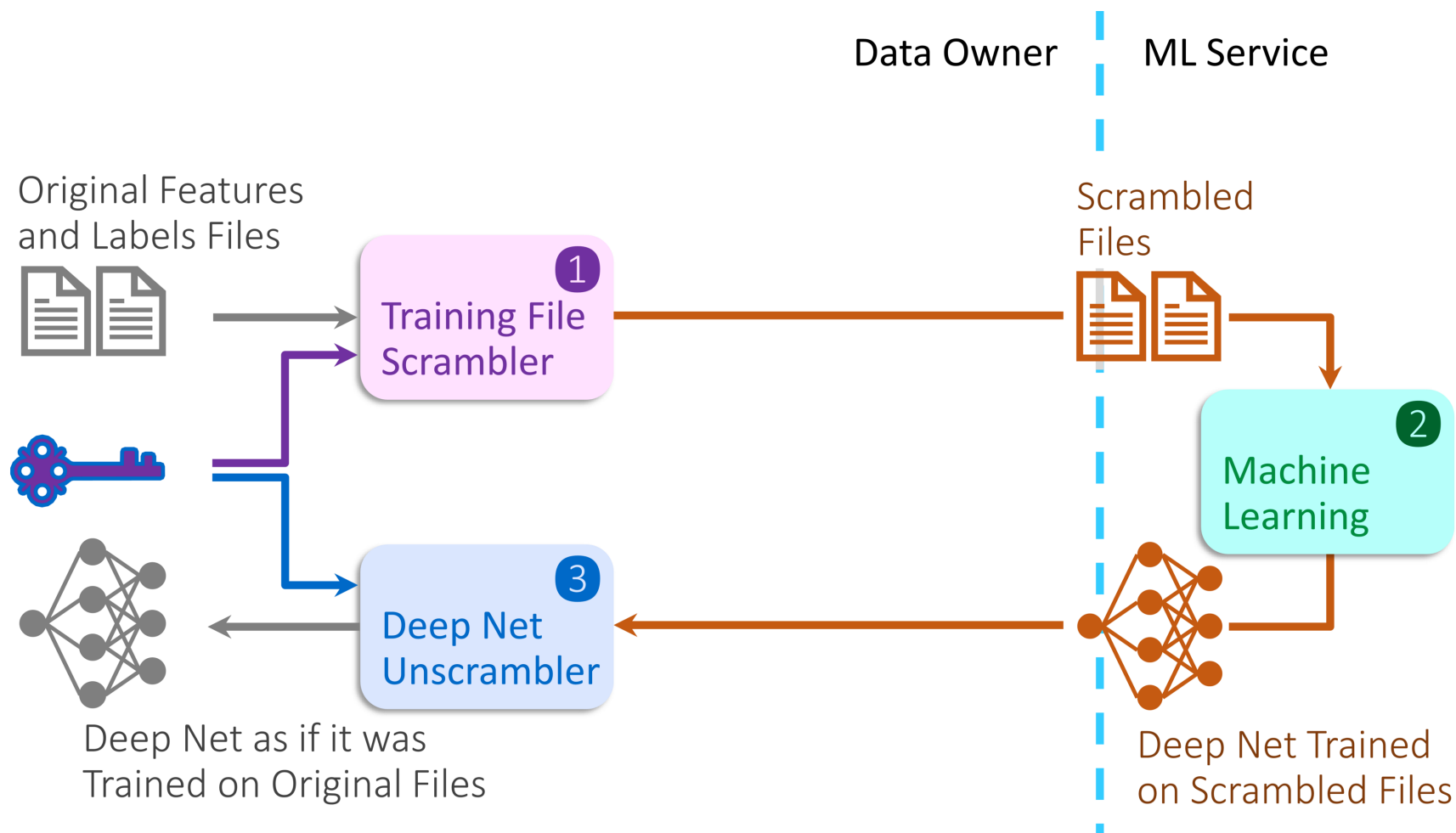
Data poisoning



Adversarial examples

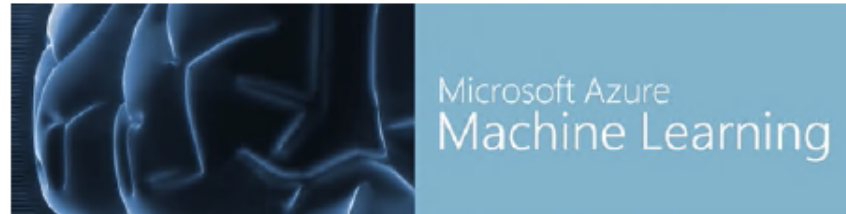
- Data confidentiality:** is the system revealing information it was not supposed to reveal?
- Adversarial examples:** feed (learning system) with malicious data in very small pieces to force misclassification.
- Data poisoning:** manipulate training data in order to control the prediction behavior of the ML system.
- System manipulation:** feed (already trained system) input data to steer it to a chosen direction.
- Transfer learning attack:** ML system is built by fine-tuning a pretrained model that is widely available.

Learning with Encrypted Data



Examples: ML-as-a-service

Amazon Machine Learning



Service	Model types / training algorithms
Amazon	Logistic regression
Google	Logistic regression, (convolutional / recurrent) neural networks, ...
Microsoft	Logistic regression, decision trees, neural networks, SVM
BigML	Logistic regression, decision trees
Algorithmia	Custom training algorithms (from third party developers)

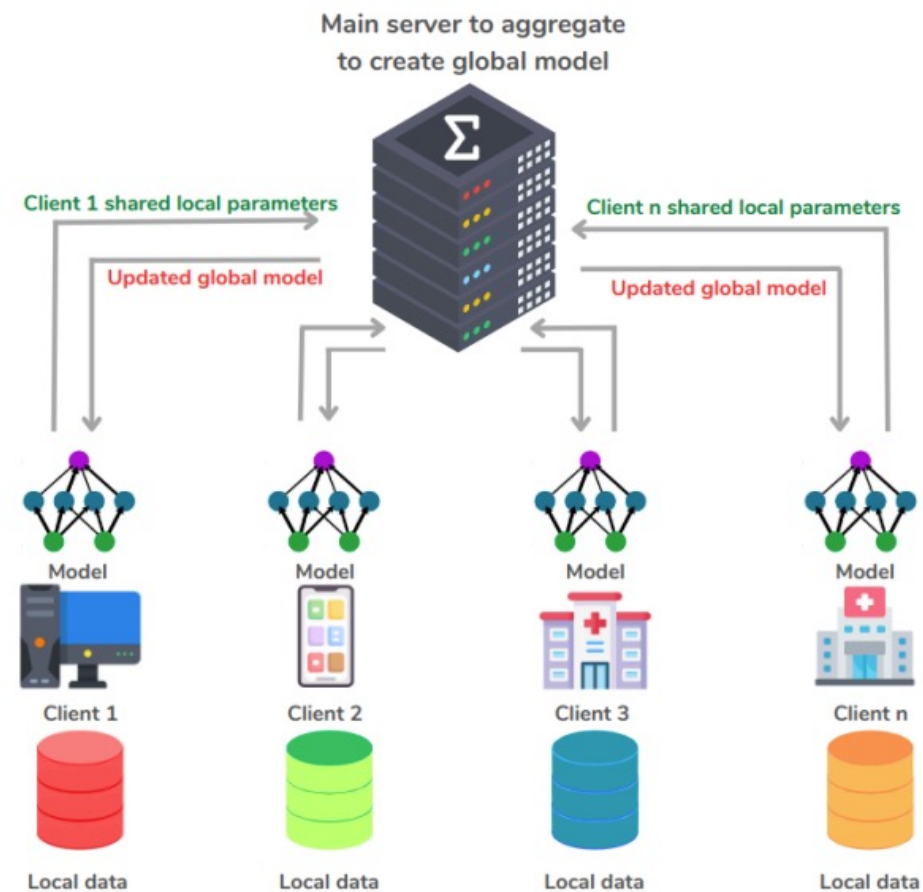
Data Privacy and Federated Learning

Are you comfortable sending your data to cloud providers?

Are you comfortable sharing your data to the community for global goods?

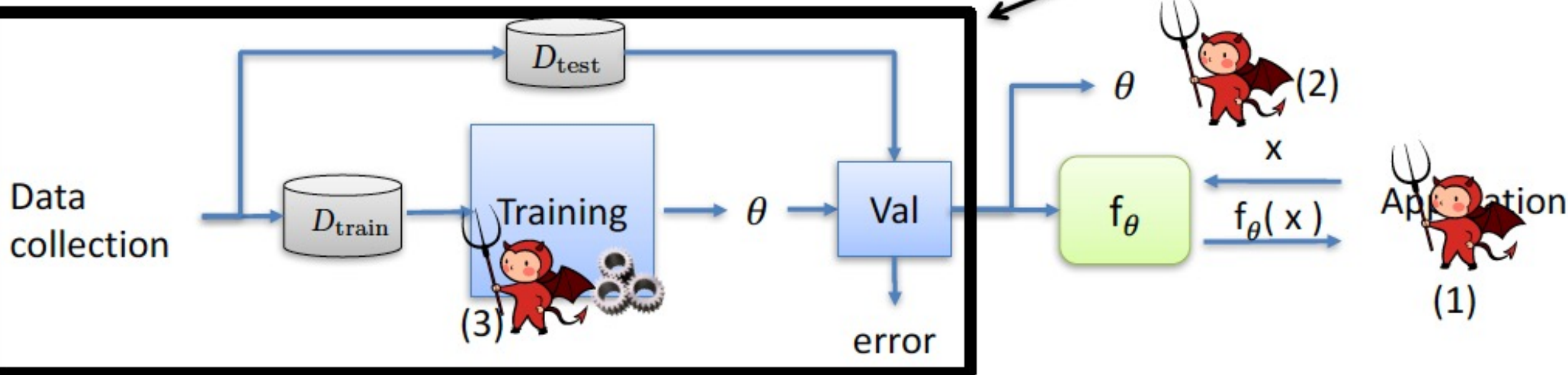
Privacy is a real issue!

Federated Learning



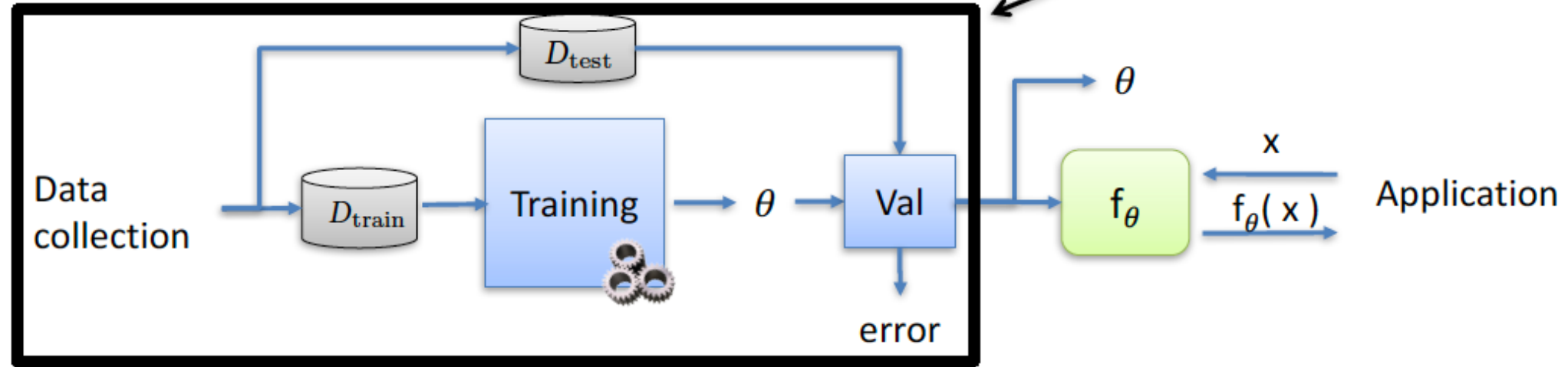
New Threat Models in ML

Isolated, secured environment



New Threat Models in ML

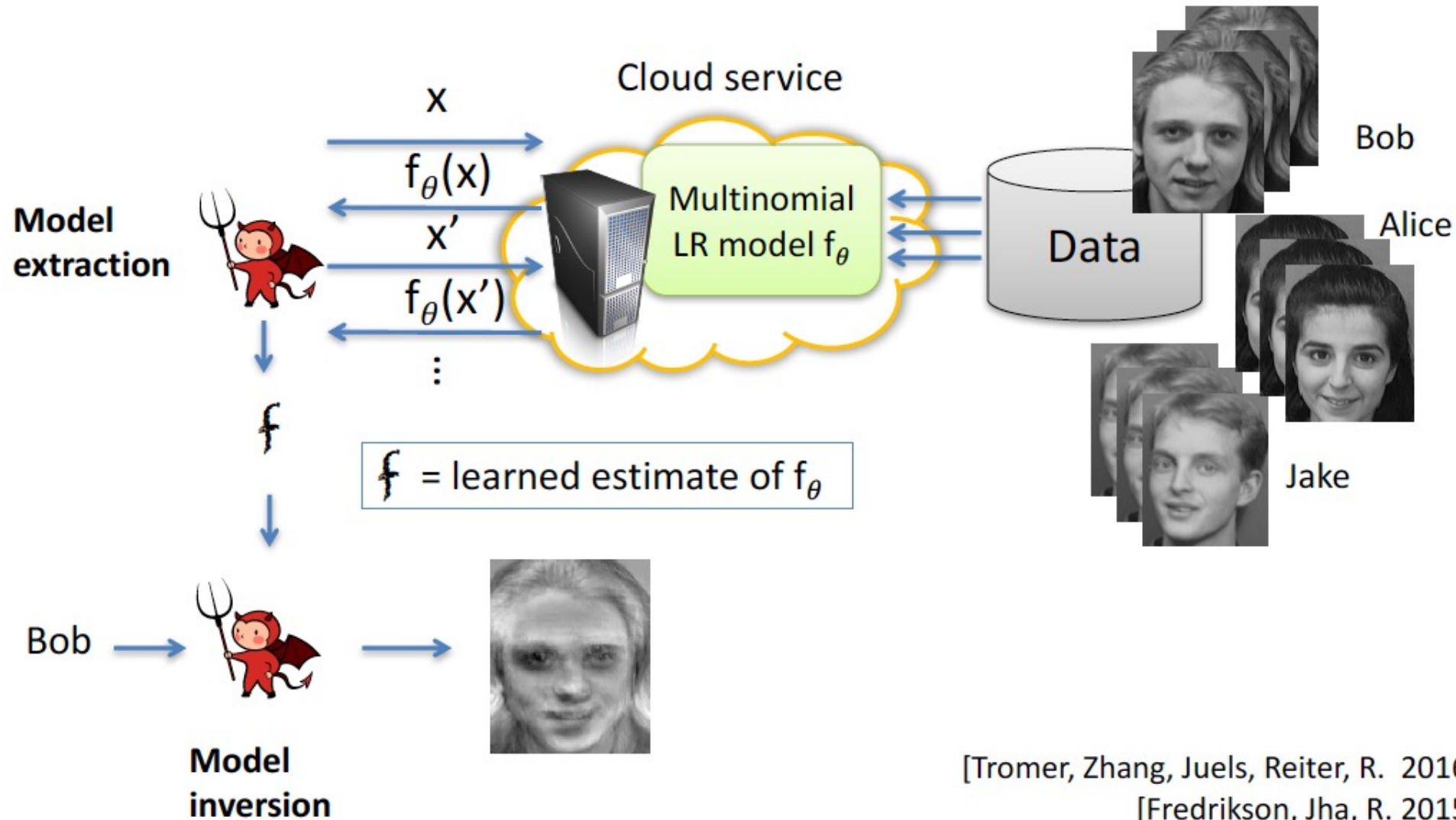
Isolated, secured environment



	Adversarial goal(s)	Adversarial abilities
(1) Model extraction	Learn θ	Query f_θ on adversarial points and see response
(2) Model inversion	Learn information about training data	Access to θ
(3) Malicious training	Exfiltrate training data through validated model θ	Specify Training algorithm, access to f_θ

Concrete example: recovering recognizable faces

Given access to facial recognition model f_θ can we reconstruct recognizable images of training set members?

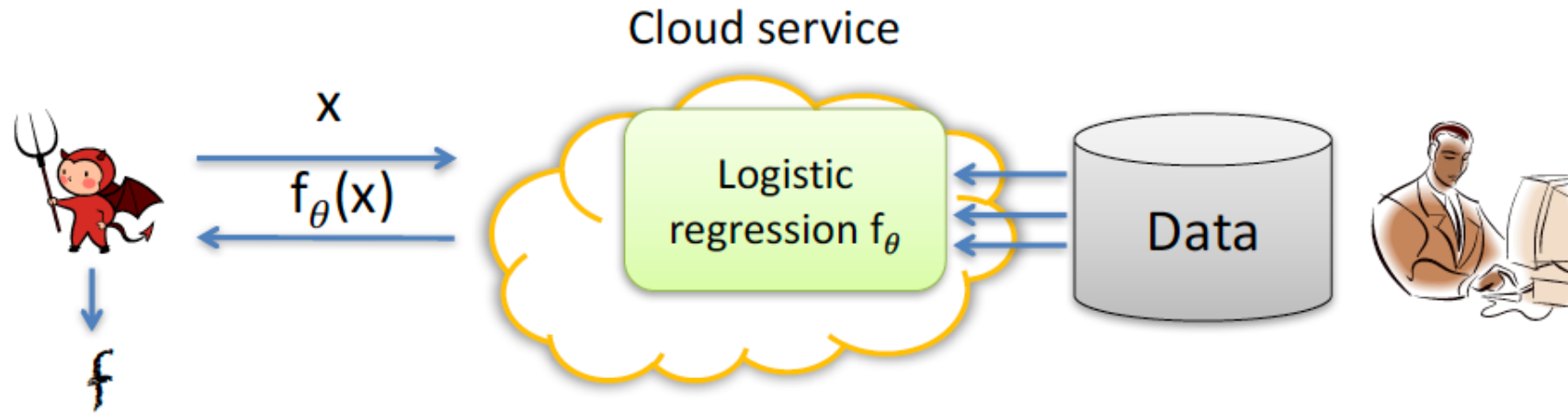


[Tromer, Zhang, Juels, Reiter, R. 2016]

[Fredrikson, Jha, R. 2015]

Model extraction attacks

Adversarial client seeks to learn close approximation of f_θ in as few queries as possible



$$f_\theta(x) = 1 / (1 + e^{-(w^*x + b)})$$

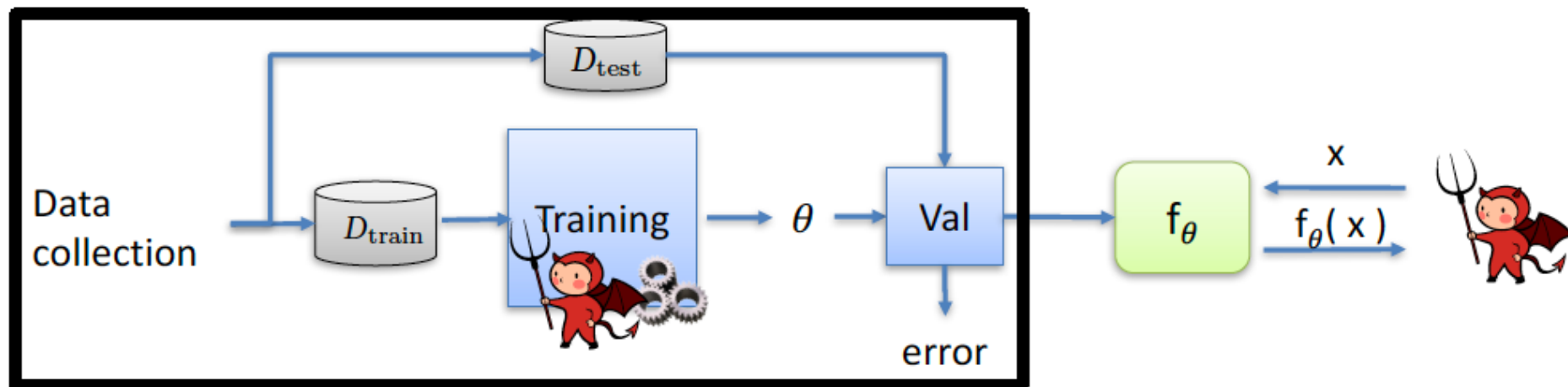
$$\ln\left(\frac{f_\theta(x)}{1 - f_\theta(x)}\right) = w^*x + b$$

Linear equation in
 $n+1$ unknowns w, b

Query $n+1$ random points $\rightarrow$ solve linear system of $n+1$ equations
 $\sim 100x$ fewer queries than [Lowd, Meek 2005]

Malicious training scenario

[Song, R., Shmatikov 2017]



Sensitive data owner somehow tricked into using a malicious algorithm in isolated environment to train deep neural network, logistic regression, etc.

ML algorithm marketplaces (e.g., Algorithmia.com)

Backdoored ML library

Can't trivially steal sensitive data by sending over network

If f_θ validates, then data owner uses it in some adversarially-accessible application

Can we modify Training algorithms so that:

- (1) Perform well on primary task
- (2) Allows recovery of training data via calls to f_θ

Questions?